

Aditya Kulkarni

adityakulkarnius@gmail.com | linkedin.com/in/aditya-kulkarni-355b81217 | kulkarniadiya.com | github.com/Aditya-k24 |

Raleigh, NC (open to relocation)

PROFESSIONAL SUMMARY

Engineer focused on agentic AI systems and LLM infrastructure. At Isomer AI, built multi-step workflow orchestration with Temporal.io and multi-model LLM routing across Claude, GPT-4o, and Gemini via MCP tooling -- 394 commits, 7 production deployments. Built AlgoPulse, an event-driven agent orchestration platform reaching 2,000 async inference requests/second with durable Temporal state machines and Kafka fault tolerance. Built a vendor-neutral LLM eval harness tracking hallucination rate, p50/p95 latency, and cost/call across 3 models with statistical significance. Python primary language across AI/ML projects. MS CS at NC State (May 2026). Portfolio: kulkarniadiya.com

CORE COMPETENCIES

Python

Agentic Workflow Orchestration

Temporal.io

LLM Pipelines (Claude, GPT-4o, Gemini)

MCP Tooling

LLM Evaluation & Observability

Kafka / Event-Driven Systems

RAG / Vector Databases

TypeScript / Node.js

Java / Spring Boot

PostgreSQL / Redis

Docker / Kubernetes / AWS

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Built Temporal.io workflow orchestration for AI-powered claims automation -- durable multi-step agent execution with fault-tolerant state machines, RBAC, and real-time event pipelines; 394 commits across 7 production deployments
- Integrated multi-model LLM routing across 3 providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models via MCP tooling; built document processing pipelines (PDF, email via Microsoft Graph API, S3) and Auth0 authentication in TypeScript, Node.js, Express, PostgreSQL, and Redis

NC State University April 2025 - Present

Graduate Research Assistant

- Designed a cloud-based genomic data platform (JBrowse 2 on Google Cloud Run) serving 500GB datasets to 100+ researchers worldwide; improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool (OCR, NLP, ML) that improved accuracy by 25%; automated deployments on AWS and GCP with Grafana monitoring; supported model red-teaming

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built distributed pipelines (Hadoop + Spark batch, Kafka streaming) for support ticket automation; reduced processing time by 15% and latency by 20%

PROJECTS

[AlgoPulse - Agentic AI Orchestration Platform](#) Temporal + Kafka

Identified API-polling bottleneck under AI inference load; re-architected to event-driven streaming. Supabase Edge Functions stream agent reasoning loops to the client in real time. Temporal.io manages durable workflow state -- agent execution survives failures, state is never lost. Kafka coordinates worker fault tolerance. Reached 2,000 async inference requests/second with zero message loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

LLM Eval Harness LLM Observability

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 3 task types on a 2,000-example frozen test set (3 repeated runs). Tracked ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, and cost/call per model. Statistical significance via paired bootstrap CIs, McNemar test, and Holm correction. JSONL + Parquet storage; DuckDB analytics; Streamlit dashboard with accuracy vs. cost scatter, latency bars, hallucination heatmap.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

Glance AI - Privacy-First RAG System Enterprise AI

Per-tenant data isolation at the vector retrieval layer for regulated environments (fintech, healthcare). Each retrieval context is scoped to its tenant boundary before LLM input -- prevents cross-tenant contamination. Deterministic output architecture ensures factual correctness for regulated data, mitigating hallucination risk.

Python, LLMs, Vector Databases, RAG

VectorLift - Semantic Search Engine Python / ML

Production-grade semantic search on MS MARCO: BM25 baseline (Elasticsearch), dense bi-encoder retrieval (sentence-transformers fine-tuned with in-batch negatives), cross-encoder re-ranker. Kafka-driven indexing into Elasticsearch and Qdrant; FastAPI serving with per-stage latency tracking; NDCG@10/MRR@10/MAP evaluation with bootstrap significance testing.

Python, PyTorch, Sentence-Transformers, Elasticsearch, FAISS, Qdrant, Kafka, FastAPI, Prometheus, Grafana

PulseRank - Real-Time Leaderboard Distributed Systems

Multi-store event-driven platform: MySQL outbox + idempotency keys, MongoDB TTL projections, Redis sorted sets for sub-ms reads; live score updates over Server-Sent Events. Multi-store consistency validated with Testcontainers; benchmarked with k6; OpenTelemetry + Prometheus observability.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

Inventory Reservation Service Java / Spring Boot

Transactional inventory reservation API in Java 21 + Spring Boot preventing overselling via PostgreSQL row-level locking across multi-SKU reservations. Concurrency correctness verified with Testcontainers: two concurrent requests produce exactly one success and one 409 Conflict. Flyway migrations, Spring Security RBAC, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Testcontainers, Docker

EDUCATION

North Carolina State University -- MS Computer Science	Aug 2024 - May 2026
Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University -- BTech Computer Science	Aug 2020 - May 2024
CGPA: 9.32/10 Machine Learning, Big Data, DS & Algorithms, Blockchain	

AWARDS & LEADERSHIP

HackNC24 Top 10 -- GenAI Finance App (PerCent); top 10 of 100+ teams, 48-hour build	2024
Sanjeevani Health App -- Top 1% of 1,100+ submissions; 87% ML classification accuracy	2023
IEEE Publication -- Mapify: T5-transformer pipeline, 60% review time reduction, 500+ attendees	2022
Google Developer Student Club Lead -- 4+ events, 500+ students	2022-2024

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, R, C	
AI/ML: MCP, LangChain, Hugging Face, PyTorch, TensorFlow, RAG, OpenAI/Anthropic/Google SDKs	
Backend: Node.js, FastAPI, Spring Boot, Express, NestJS, REST APIs	Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant
Infrastructure: AWS, GCP, Docker, Kubernetes, Kafka, Temporal.io, Terraform, Spark	Observability: Prometheus, Grafana, Sentry, OpenTelemetry
DevOps: GitHub Actions, Docker Compose, Testcontainers, Flyway	